BIOGAS PROJECT

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Biogas Project Report: Biogas Production from Dung

Introduction

Biogas production through anaerobic digestion is an efficient and sustainable method of converting organic waste, such as dung, into renewable energy. In this project, biogas is produced from dung placed in a sealed container. A gas pipe is connected to the container to collect the produced biogas, which is then stored in a tyre tube system for future use. This report outlines the setup, process, and key factors involved in the biogas production project.

Materials and Setup

1. **Container for Dung**

- A container made of plastic or metal was used to store the dung. The container was kept airtight to prevent oxygen from entering, creating an anaerobic environment necessary for the digestion process.

2. **Gas Pipe System**

- A gas pipe was securely connected to the lid of the dung container using a strong adhesive, magic glue. This ensured that the pipe remains leak-free and directs the produced biogas away from the container.
- The other end of the pipe was connected to a three-outlet system, which divided the biogas flow into different pathways.

3. **Tyre Tubes for Gas Storage**

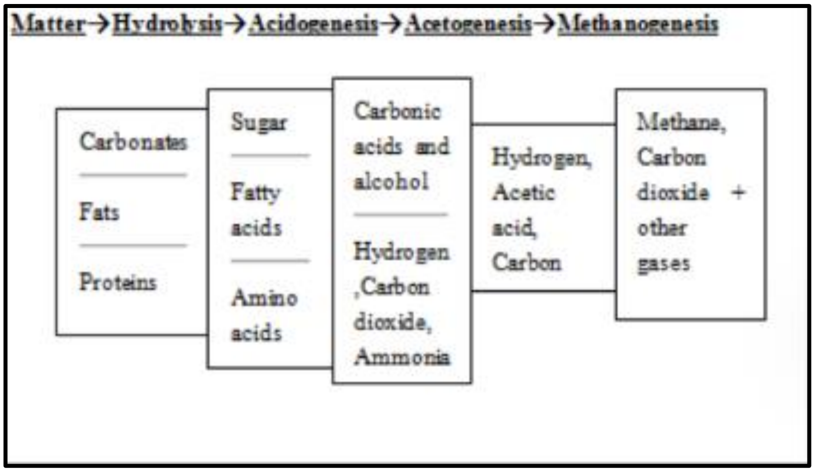
- An outlet of the pipe was connected to the tyre tube. This tyre tube functioned as gas storage tank, accumulating the biogas produced during digestion.

4. **Gas Release Outlet**

- An outlet was left open, which allowed gas to be used immediately. This outlet can be connected to a cooking stove or other equipment that uses biogas for energy.

Process of Biogas Production

Anaerobic digestion is a multi-stage biological and chemical process that occurs in four key phases: hydrolysis, acidogenesis, acetogenesis, and methanogenesis. These stages, depicted in Figure 1, involve different groups of microorganisms, some of which work together in a syntrophic relationship, each with specific environmental needs. During hydrolysis and fermentation, microorganisms break down complex polymers and monomers, producing acetate, hydrogen, and various volatile fatty acids like propionate and butyrate. Hydrolytic microbes secrete enzymes such as cellulase, xylanase, amylase, lipase, and protease, and include strict anaerobes like *Bacteroides*, *Clostridia*, and *Bifidobacteria*, as well as some facultative anaerobes like *Enterobacteriaceae* and *Streptococci*. Higher volatile fatty acids are then converted into acetate and hydrogen by acetogenic bacteria. However, the accumulation of hydrogen can inhibit acetogenic bacteria, so maintaining a very low hydrogen partial pressure is crucial. Although much of the microbial network in the methanogenic phase is not fully understood, it is believed that hydrogen plays a key role in the metabolism of methanogens. Finally, methanogenic bacteria produce methane from acetate or hydrogen and carbon dioxide, completing the anaerobic fermentation process.



Hydrolysis

Biomass typically consists of large organic polymers like proteins, fats, and carbohydrates. These complex molecules are broken down into smaller components such as amino acids, fatty acids, and simple sugars during hydrolysis. This crucial first step in anaerobic digestion is carried out by fermentative bacteria that convert the complex organic matter into soluble molecules. Some of the products from hydrolysis, like hydrogen and acetate, can later be utilized by methanogens in the subsequent stages of digestion. The remaining relatively large molecules must undergo further breakdown in the acidogenesis stage before they can contribute to methane production.

Acidogenesis

After hydrolysis, acidogenesis follows, where acidogenic microorganisms continue to decompose the biomass and organic compounds. These fermentative bacteria generate an acidic environment in the digester, producing byproducts such as ammonia, hydrogen (H_2), carbon dioxide (CO_2), hydrogen sulfide (H_2S), and shorter volatile fatty acids like acetic, propionic, and butyric acids. These products are crucial intermediates for the later stages of anaerobic digestion.

Acetogenesis:

In acetogenesis, acetogens convert the organic products from acidogenesis into acetate, acetic acid, CO_2, and hydrogen. This process prepares the remaining biomass for methanogens to utilize in the production of methane. Essentially, acetogenesis makes the breakdown products more suitable for methanogenic bacteria to complete the digestion cycle.

Methanogenesis:

The final stage of anaerobic digestion, methanogenesis, involves methanogenic microorganisms converting acetate and carbon dioxide, two key products from the previous stages, into methane. There are two primary pathways for methane formation: one involves the reduction of CO_2 using hydrogen ($CO_2 + 4H_2 \rightarrow CH_4 + 2H_2O$), while the other involves the conversion of acetic acid into methane and carbon dioxide ($CH_3COOH \rightarrow CH_4 + CO_2$). Although both pathways contribute, the conversion of acetic acid is the predominant route for methane production. This stage results in the generation of methane and carbon dioxide, the main outputs of anaerobic digestion.

Principle of gas production

The principle of gas production is based on the decomposition of organic or biological materials, during which various gases are released. Organic decomposition can occur through two primary pathways:

1. **Aerobic Decomposition**: This process takes place in the presence of oxygen. During aerobic decomposition, microorganisms break down organic matter, and the products

typically include carbon dioxide (CO_2), water (H_2O), and energy. This type of decomposition is often faster and more efficient than anaerobic decomposition due to the availability of oxygen.

2. **Anaerobic Decomposition**: This occurs in the absence of oxygen. Anaerobic decomposition is carried out by anaerobic microorganisms that break down organic matter into simpler compounds, leading to the production of gases such as methane (CH_4), carbon dioxide (CO_2), and other trace gases like hydrogen sulfide (H_2S). Anaerobic decomposition is slower than aerobic decomposition but is essential for processes like biogas production, where methane is captured as a valuable energy source.

Both decomposition processes release gases, but the types of gases produced and the efficiency of the process vary depending on the presence or absence of oxygen.

Composition of Biogas

The primary gases produced in biogas are methane (CH_4) and carbon dioxide (CO_2). Biogas is approximately 20% lighter than air and has an ignition temperature ranging from 50°C to 750°C. It is a colourless and odorless gas that burns with a clear blue flame, similar to LPG (liquefied petroleum gas). The calorific value of 1 cubic meter of biogas is about 22 MJ when burned at 60% efficiency.

The composition of biogas can vary depending on the raw materials used in its production, but a typical composition is outlined in the table below:

Matter	Percentage
Methane, CH_4	50-75
Carbon dioxide, CO_2	25-50
Nitrogen, N_2	0-10
Hydrogen, H_2	0-1
Hydrogen sulfide, H_2S	Traces
Water vapour	Traces
Oxygen, O_2	0-2

Note that the percentages may fluctuate based on factors like the type of organic material (e.g., animal waste, food waste, or agricultural residues) used in the anaerobic digestion process.

Factors Affecting Gas Production

The volume of gas produced from bioresources in anaerobic digestion depends on several factors. Key factors include:

- **Temperature**: Temperature plays a crucial role in optimizing the stability and efficiency of the anaerobic digestion process. There are no strict standards, but maintaining a narrow temperature range is vital for optimal digestion:
  - **Mesophilic conditions**: Digestion proceeds best at **30 - 40°C**.
  - **Thermophilic conditions**: The optimum range is **50 - 60°C**.

The choice of temperature is influenced by climate. In warm climates, digesters can operate without additional heating, often utilizing soil for insulation or a greenhouse covering. In colder climates, heating may be necessary, and this can be achieved at lower costs by using natural materials like leaves, sawdust, or straw to compost around the digester. For ideal fermentation, the temperature should generally be kept above **30°C**.

- **pH**:
  The pH level significantly impacts gas production. A low pH inhibits the growth of methanogenic bacteria, thereby reducing gas generation. Efficient digestion occurs at a

near-neutral pH, within the range of **6.0 - 8.0**. A slightly alkaline pH suggests that pH fluctuations are not extreme. If the pH becomes too low, it can be corrected by dilution or the addition of lime.

- **Nutrients**

  Optimal microbiological activity is crucial for efficient gas production, and this is heavily influenced by nutrient availability. The most important nutrients for microbial growth are **carbon (C)** and **nitrogen (N)**. The **C/N ratio** is critical in selecting raw materials:
  - **N-rich materials** like domestic sewage and animal waste provide nitrogen for the growth of anaerobic organisms.
  - **C-poor materials** such as agricultural waste or green grass are rich in carbohydrates, which are vital for gas production.

Excess nitrogen can lead to the formation of ammonia (NH3), which inhibits microbial growth. Ammonia toxicity can be mitigated by reducing loading rates or diluting the mixture. Ideally, a **C/N ratio of 30:1** by weight should be maintained for optimal digestion. This ratio can be adjusted by combining high-carbon materials with nitrogen-rich materials.

- **Toxic Materials**

Biodegradable residues often contain pollutants that can hinder anaerobic digestion. **Ammonia toxicity** can be addressed by adjusting the C/N ratio through the addition of materials like shredded bagasse or straw. Common toxic substances that may inhibit digestion include:

  - **Heavy metals** such as copper, zinc, nickel, mercury, and chromium.
  - **Soluble salts** of sodium, potassium, calcium, and magnesium, which can be either stimulatory or toxic depending on the cation.
  - **Pesticides and synthetic detergents**, which can also disrupt the digestion process.

Managing these factors effectively ensures the efficient production of biogas from organic materials.

CONCLUSION

In conclusion, maintaining an optimal pH, nutrient balance, and controlling toxic substances are essential for efficient anaerobic digestion and biogas production. A near-neutral pH, a proper C/N ratio, and careful management of toxic materials like heavy metals and ammonia ensure the stability of microbial activity, promoting high gas yields. By managing these factors, biogas production can be maximized, making it a sustainable and reliable source of renewable energy.